

E-COMMERCE SALES & CUSTOMER ANALYTICS SYSTEM

Final Business Analytics Report

Data Analytics Internship Project

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Role: Data Analytics Intern

Tools: Excel | PostgreSQL | Python | Pandas | Matplotlib | Seaborn | Power BI

Project Duration: 15 Working Days

Organization: SATRI

1. Executive Summary

This project analyzes e-commerce sales and customer data to understand sales performance, profitability, customer value, product performance, regional performance, time trends, and the relationship between discounts and profit. The analysis follows a complete analytics workflow using Excel, PostgreSQL, Python/Pandas, and Power BI.

The dataset contains 9,994 records covering orders from January 2014 to December 2017, with 793 unique customers and 1,862 unique products. Overall sales were \$2,297,200.86 and total profit was \$286,397.02, giving an overall profit margin of approximately 12.47%. There were 5,009 distinct orders and the average order value was approximately \$458.61.

The analysis identified meaningful differences in profitability across categories and regions. Technology generated the highest category sales and profit, while Furniture produced a much lower profit margin. The West region generated the highest sales and profit among the four regions. Higher discount levels were associated with substantially lower profitability in the historical data, including negative aggregate profit at several high-discount levels.

2. Business Background

The project simulates the role of a Junior Data Analyst working for an e-commerce business. The purpose of the analysis is to transform raw business data into reliable information that can support management decisions. The business needs visibility into revenue, profit, customers, products, regions, trends, and discount performance.

3. Problem Statement

Management needs to understand what is driving sales and profit, which customers provide greater value, which products and regions perform poorly, how discount levels relate to profitability, and what actions can be considered based on the available data.

4. Project Objectives

- Analyze overall sales and profitability.
- Identify top- and bottom-performing products and categories.
- Analyze customer behavior and customer value.
- Evaluate regional sales and profit performance.
- Analyze sales trends over time.
- Evaluate the relationship between discounts, sales, and profit.
- Develop an interactive Power BI dashboard.
- Produce evidence-based business insights and recommendations.

5. Dataset & Data Understanding

The cleaned dataset contains 9,994 rows and 21 columns. The main analytical fields are:

Field	Business Meaning
Order ID	Unique order identifier
Order Date	Purchase date
Customer ID	Customer identifier
Product / Product Name	Product purchased
Category / Sub-Category	Product classification
Quantity	Units sold
Sales	Revenue from the order line
Discount	Discount applied
Profit	Profit from the order line
Region	Geographic region

The data covers four years from 2014-01-03 to 2017-12-30. It contains 793 unique customers, 1,862 unique products, 3 categories, 17 sub-categories, and 4 regions.

6. Data Cleaning & Quality Assessment

Data quality checks were performed before analysis. The dataset contained no missing values, no duplicate full rows, and no invalid order/ship date relationships. Sales and quantity values were positive, and discount values were within the expected 0–1 range.

Text fields were checked for leading/trailing whitespace, date fields were converted to proper date values, and numeric fields were validated. Potential statistical outliers identified using the IQR method were not automatically removed because they may represent genuine high-value transactions. They were documented for later analytical consideration.

7. Methodology

The project used a structured analytics workflow. First, the raw dataset was profiled and its structure, quality, and business meaning were documented. The data was then cleaned and validated in Excel. A PostgreSQL database was created for SQL-based analysis. Python/Pandas was used for exploration, transformation, statistical analysis, and visualization. Finally, Power BI was used to convert the analytical results into an interactive management dashboard.

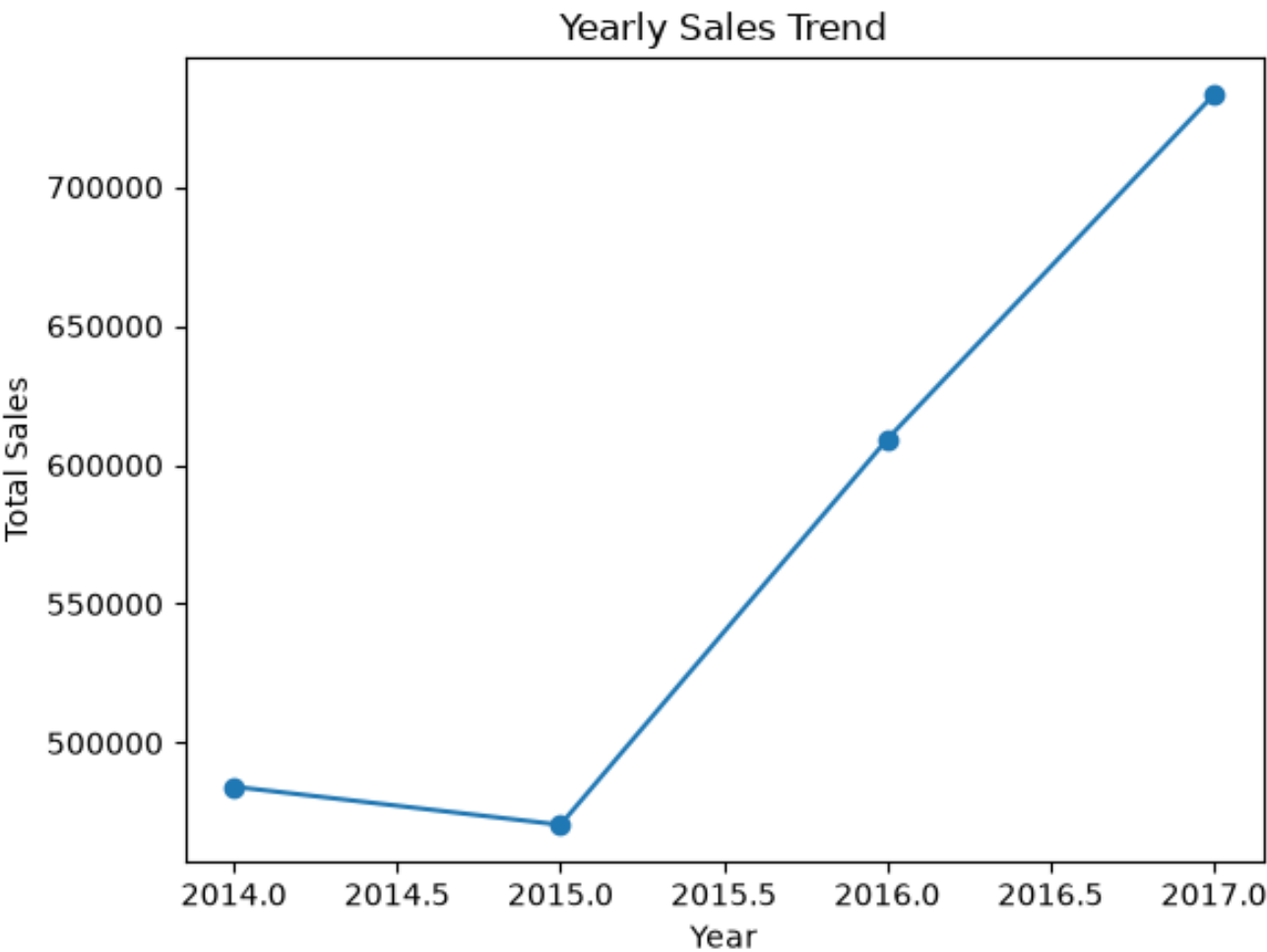
Stage	Tool	Main Activities
Data Understanding	Excel / Python	Structure, quality checks, descriptive statistics
Data Cleaning	Excel / Python	Validation, standardization, date and numeric checks
Database Analysis	PostgreSQL	Aggregations, filtering, grouping and business queries
EDA & Statistics	Python/Pandas	Visualizations, distributions, correlations and comparisons
Business Reporting	Power BI	KPIs, trends, category, product, customer and regional views

8. Sales Analysis

Overall sales performance:

KPI	Value
Total Sales	\$2,297,200.86
Total Profit	\$286,397.02
Total Orders	5,009
Total Customers	793
Average Order Value	\$458.61
Profit Margin	12.47%

Annual sales increased from approximately \$484,247.60 in 2014 to \$733,215.20 in 2017. This shows an overall upward sales trend across the four-year period.

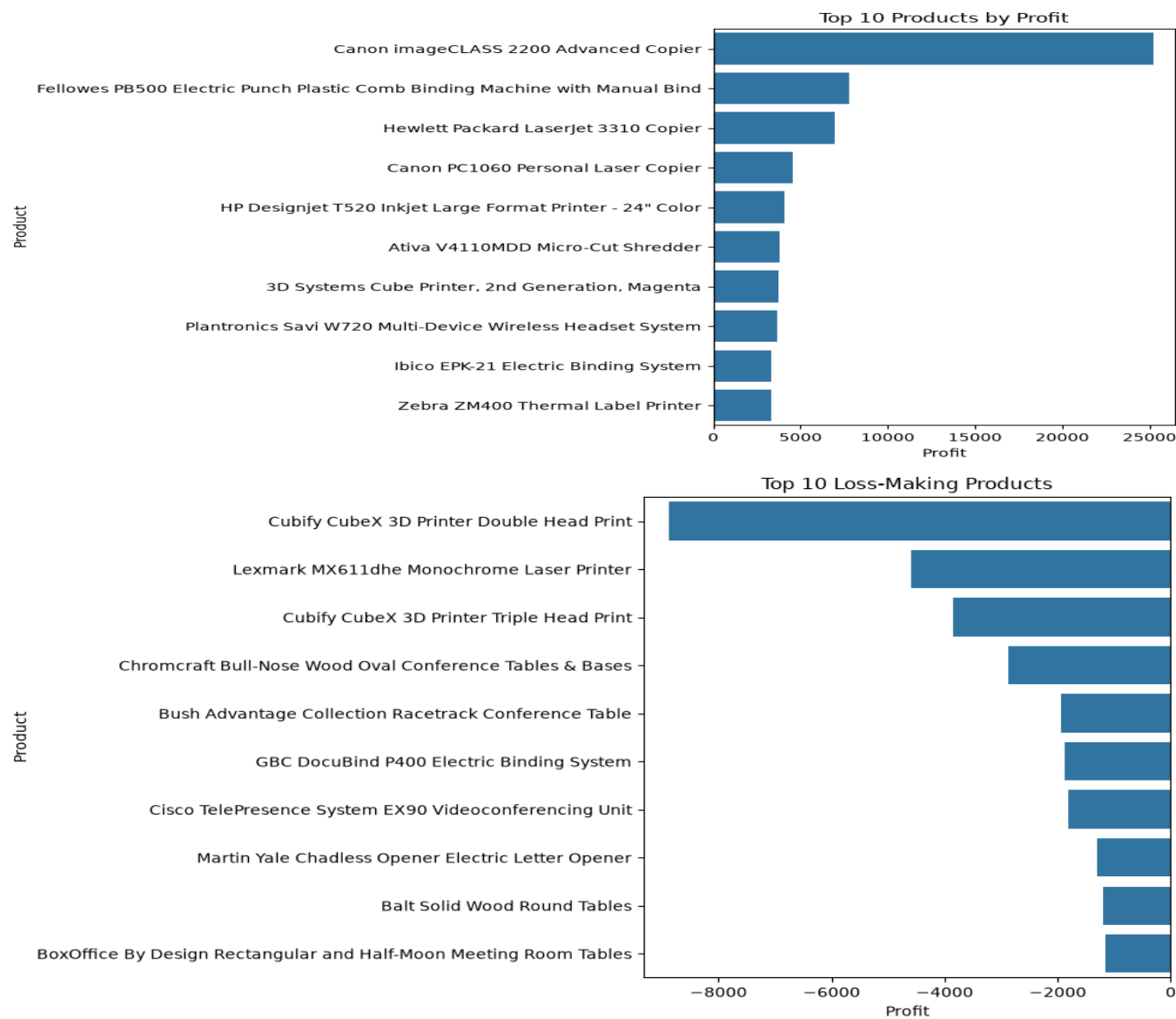


9. Product Analysis

Product-level analysis compared sales, profit, quantity, and order frequency. The Canon imageCLASS 2200 Advanced Copier was the highest-selling product and also generated the highest profit among the listed products. However, high sales did not always correspond to high profitability. For example, the Cisco TelePresence System EX90 generated approximately \$22.6K in sales but recorded a loss of approximately \$1.81K.

Examples of loss-making products included:

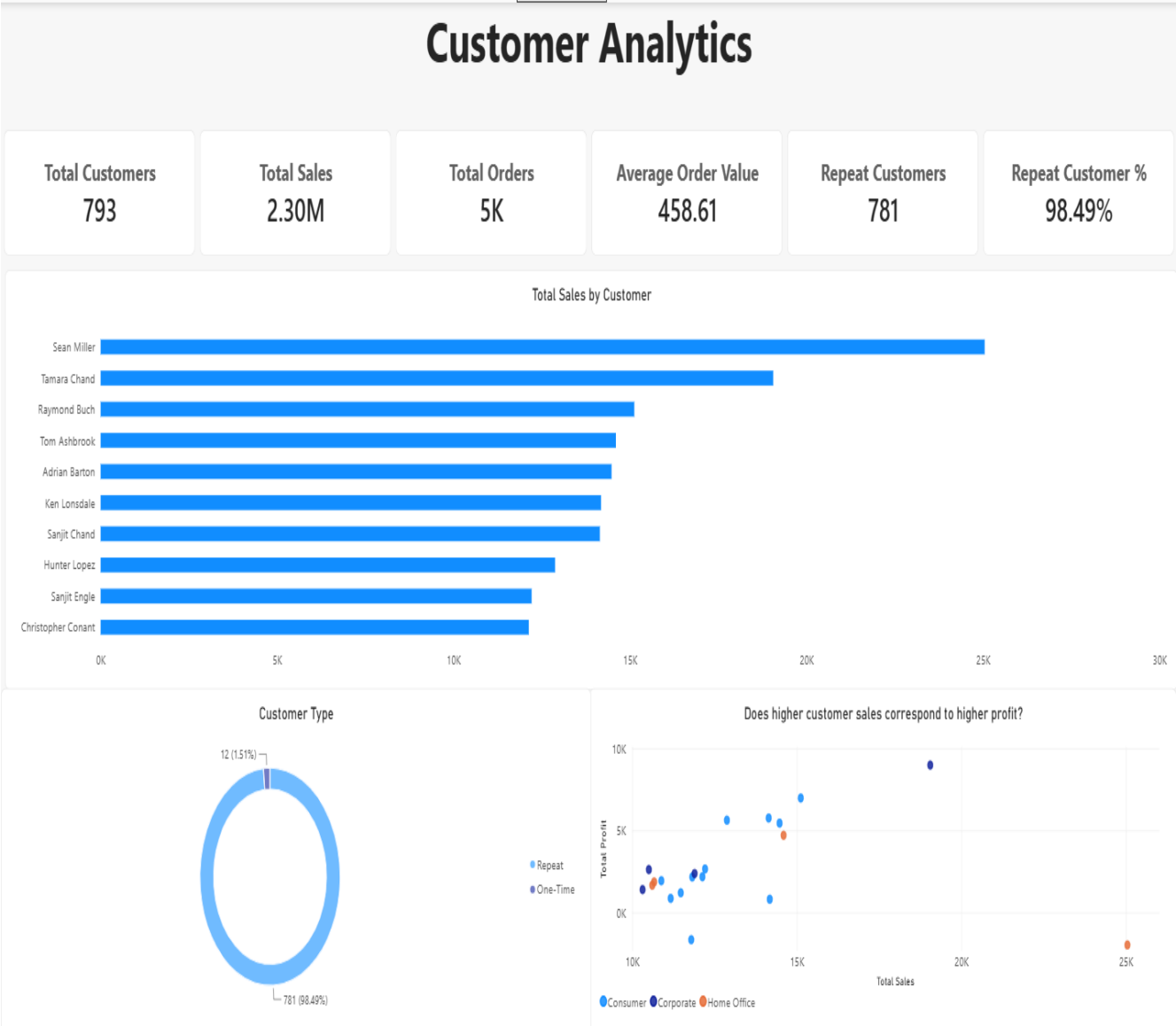
- Cubify CubeX 3D Printer Double Head Print: approximately -\$8,879.97 profit.
- Lexmark MX611dhe Monochrome Laser Printer: approximately -\$4,589.97 profit.
- Cubify CubeX 3D Printer Triple Head Print: approximately -\$3,839.99 profit.
- Chromcraft Bull-Nose Wood Oval Conference Tables & Bases: approximately -\$2,876.11 profit.



10. Customer Analysis

There were 793 unique customers. Based on first purchase year, 595 customers first purchased in 2014, 136 in 2015, 51 in 2016, and 11 in 2017. The analysis identified 781 repeat customers and 12 one-time customers.

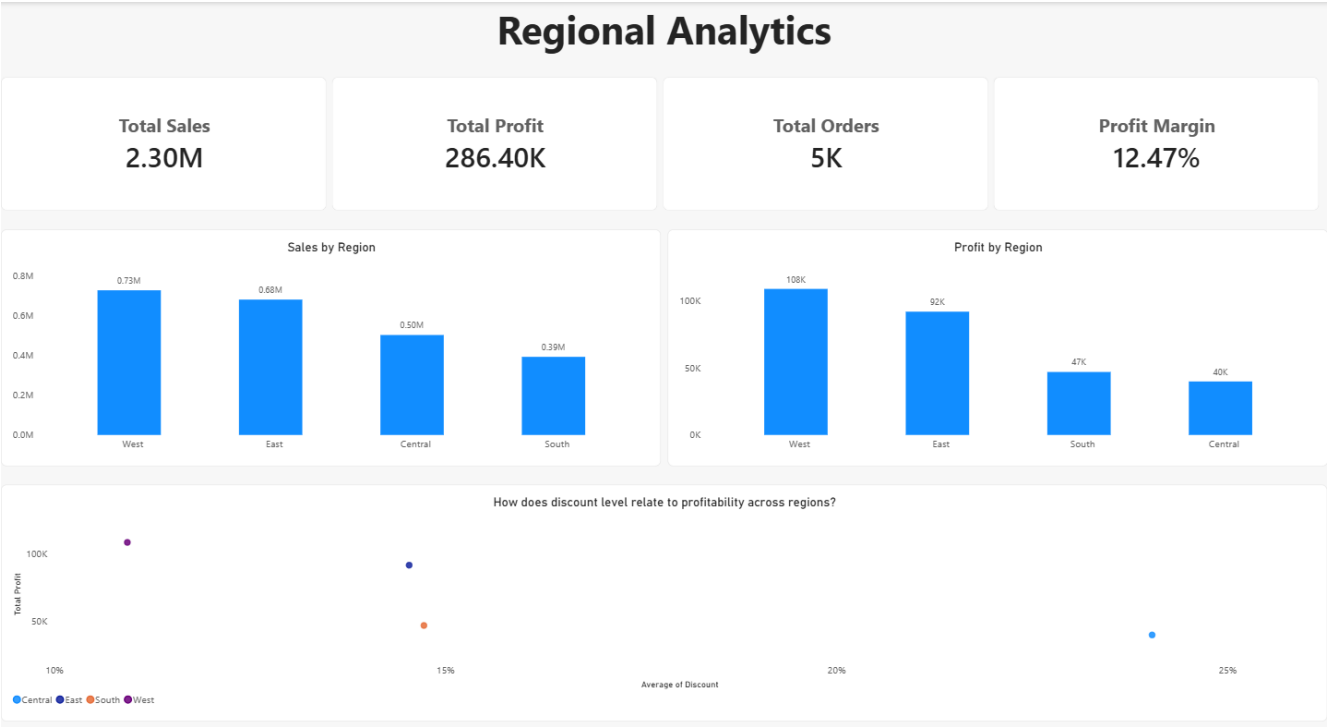
Average customer spend was approximately \$2,896.85. Customer-level analysis also showed that high sales value does not necessarily imply high profitability. For example, Sean Miller was the highest customer by sales at approximately \$25,043 but recorded a loss of approximately \$1,981.



11. Regional Analysis

Region	Sales	Profit	Profit Margin
West	\$725,457.82	\$108,418.45	14.94%
East	\$678,781.24	\$91,522.78	13.48%
Central	\$501,239.89	\$39,706.36	7.92%
South	\$391,721.91	\$46,749.43	11.93%

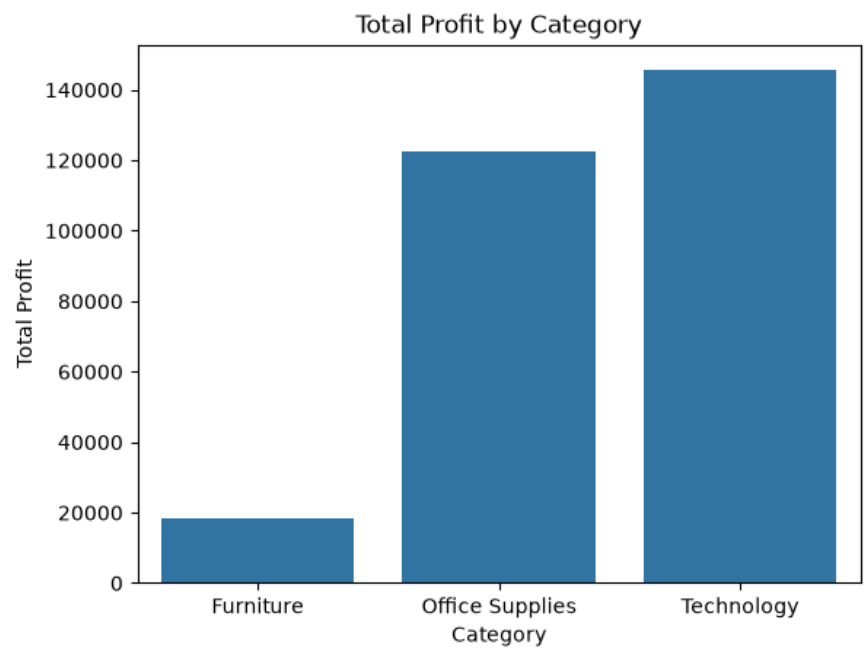
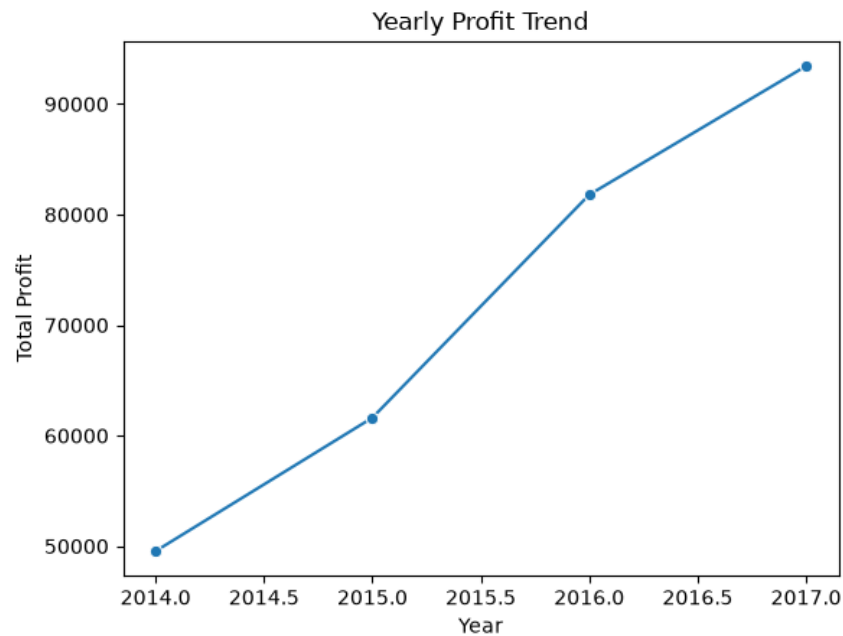
The West region recorded the highest sales and profit. Central had the lowest profit margin among the four regions, indicating a need for closer examination of its product mix, pricing, and discount patterns.



12. Profitability Analysis

Category	Sales	Profit	Profit Margin
Technology	\$836,154.10	\$145,455.70	17.39%
Office Supplies	\$719,047.00	\$122,490.90	17.03%
Furniture	\$741,999.80	\$18,451.25	2.49%

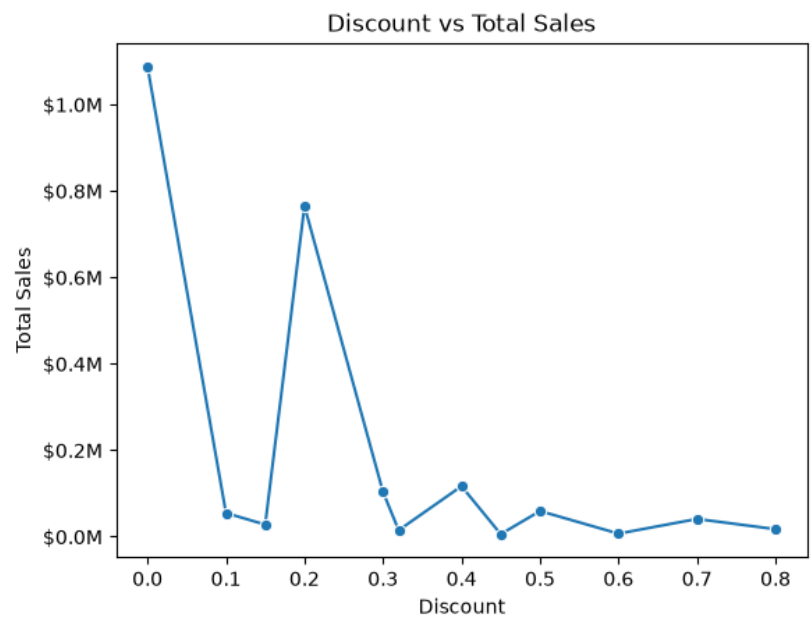
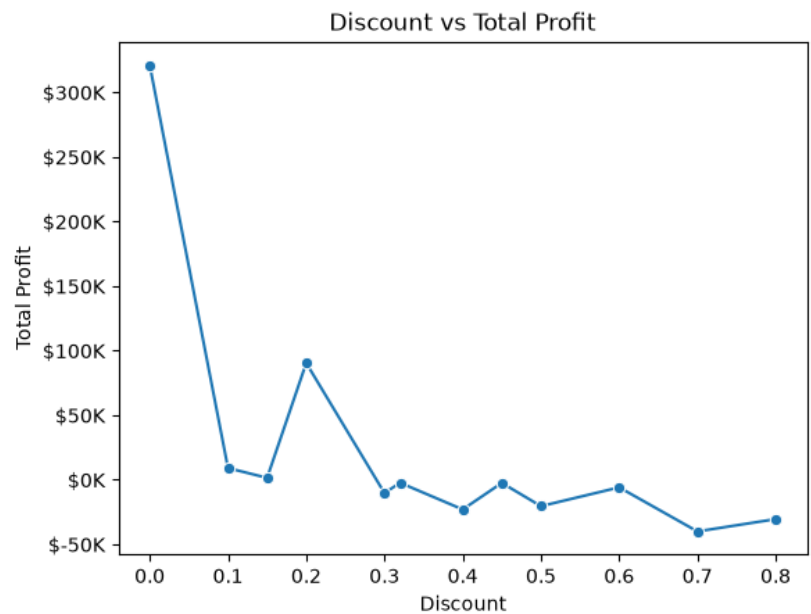
Technology generated the highest sales and profit, while Furniture generated substantial sales but a much lower profit margin. This difference demonstrates why sales alone should not be used as the only measure of business performance.



13. Discount Analysis

Discount levels were compared with aggregate sales and profit. The 0% discount group generated approximately \$320,987.60 profit, while several high-discount groups recorded negative aggregate profit. For example, the 70% discount group recorded approximately -\$40,075.36 profit and the 80% group approximately -\$30,539.04.

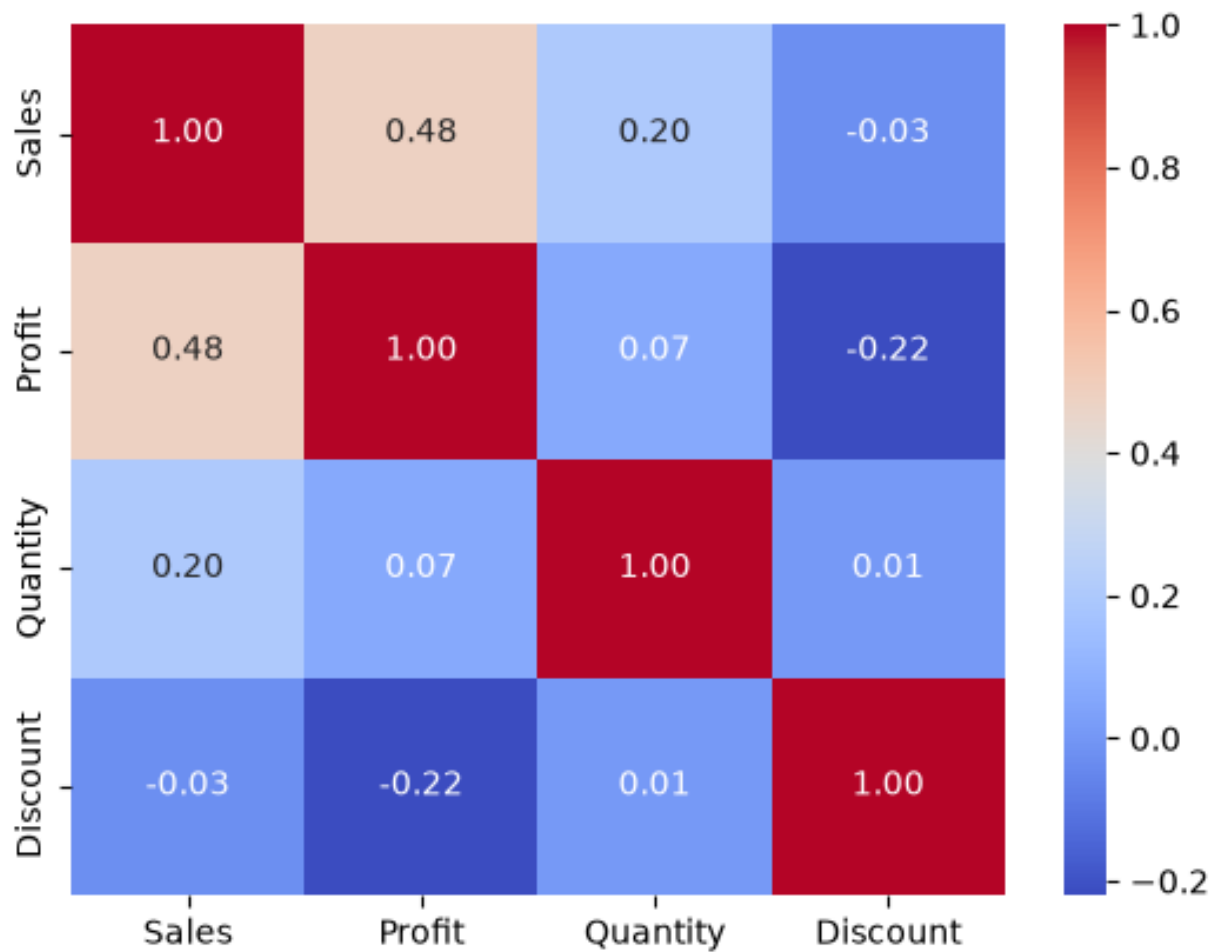
The analysis indicates an association between higher discount levels and weaker profitability in this historical dataset. This should not be interpreted as proof that discounts directly caused the losses, because other factors may also affect profit.



14. Statistical & Correlation Analysis

Metric	Sales	Profit	Quantity	Discount
Mean	229.858	28.657	3.790	0.156
Median	54.49	8.667	3	0.20
Std. Dev.	623.245	234.260	2.225	0.206
Minimum	0.444	-6599.978	1	0
Maximum	22638.48	8399.976	14	0.80

Sales and profit distributions are right-skewed, as the means are substantially higher than their medians. The correlation analysis showed a strong positive relationship between sales and profit, a weak positive relationship between quantity and sales, and a negative relationship between discount and profit. Correlation measures association and does not establish causation.

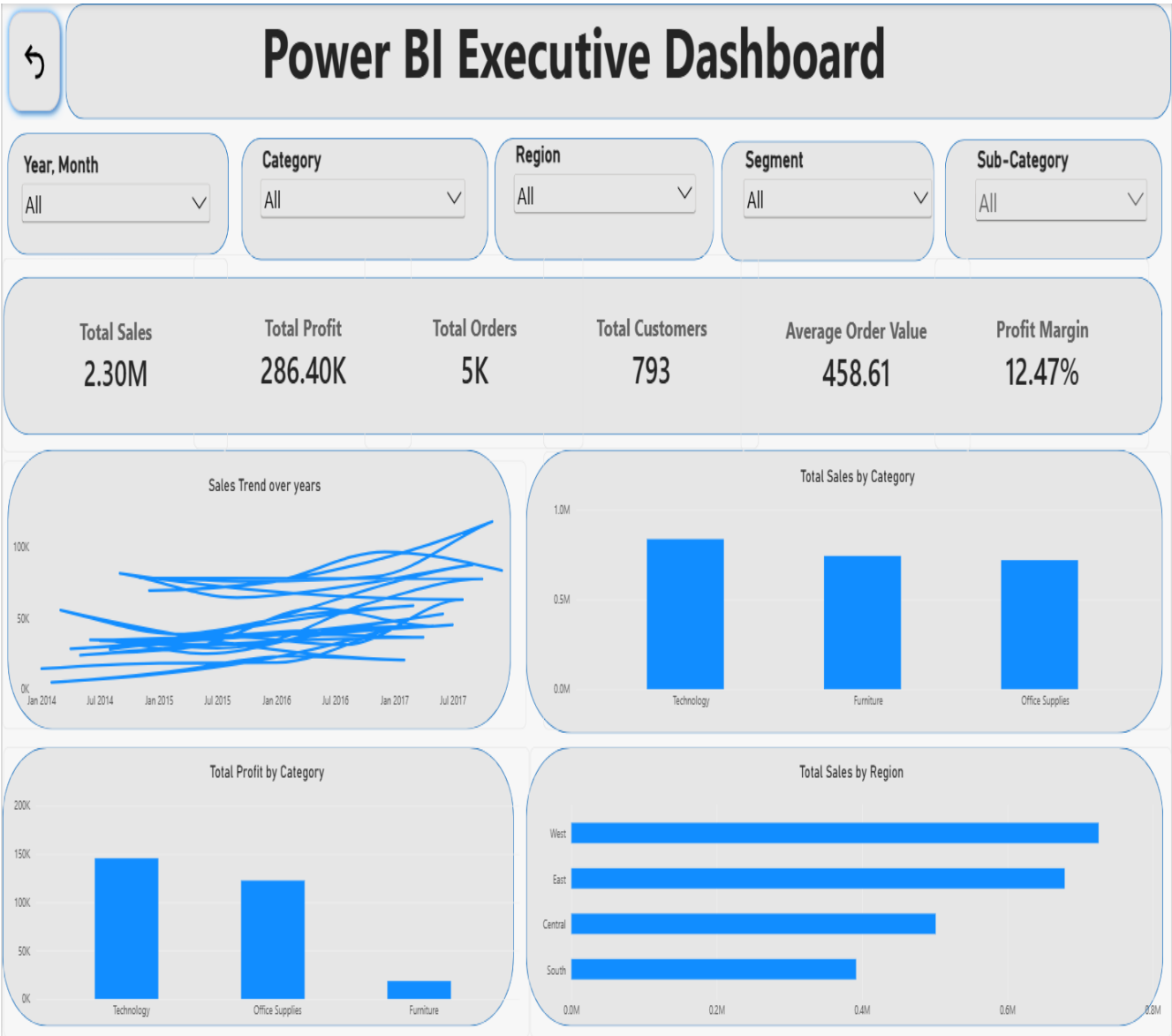


15. Power BI Dashboard

Power BI was used to present the analytical results in an interactive dashboard for management. The dashboard is designed to provide a clear overview of business performance and allow users to explore sales, profitability, customers, products, regions, and discount-related performance.

15.1 Executive Dashboard

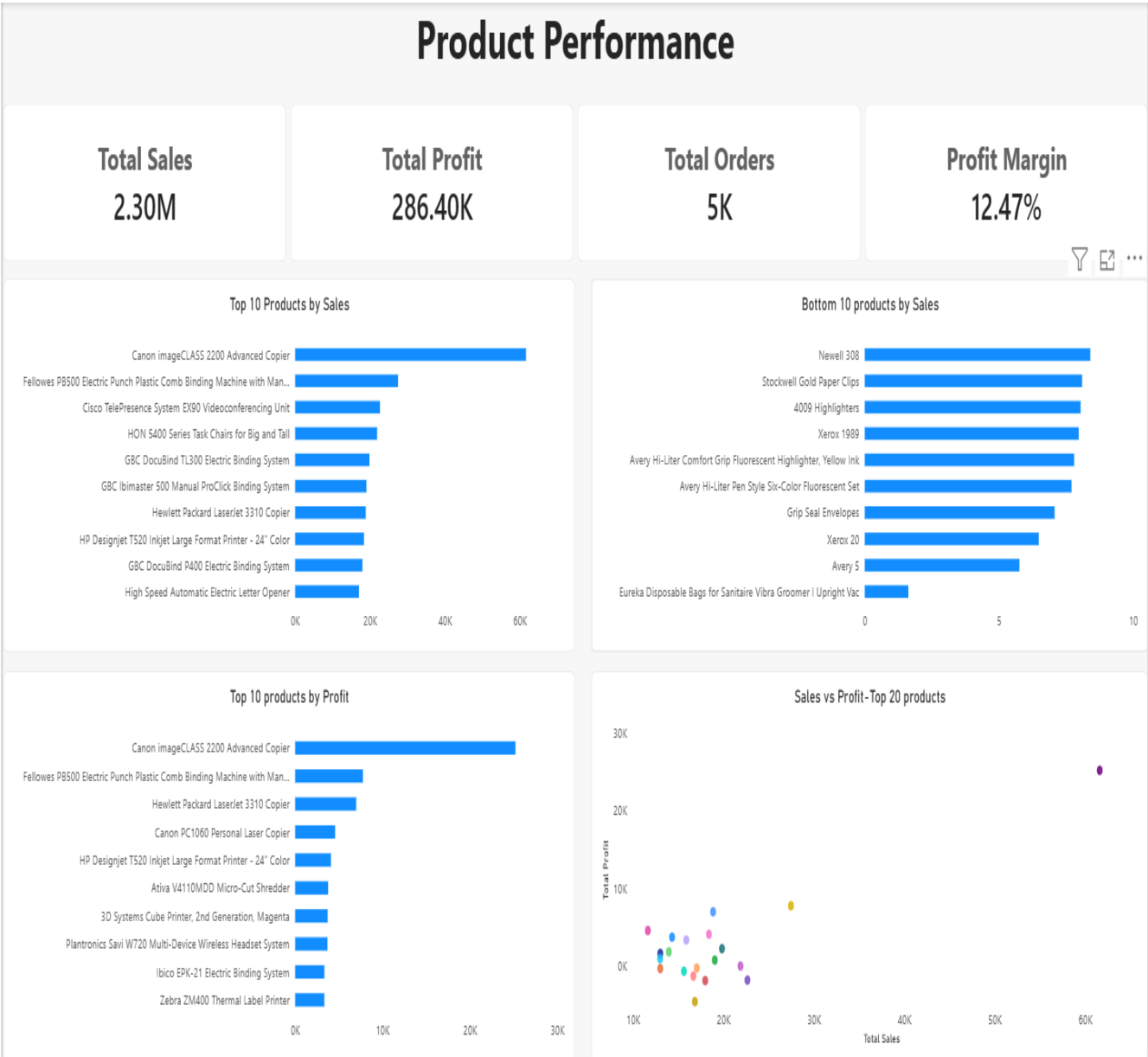
The Executive Dashboard presents Total Sales, Total Profit, Total Orders, Total Customers, Average Order Value, and Profit Margin, together with sales trends, category performance, and regional performance.



Dashboard interpretation: The Executive Dashboard provides an overview of the company's overall sales and profitability performance. Total Sales are **\$2.30 million**, with **\$286.40K total profit** and an overall **profit margin of 12.47%**. The dashboard also shows **5,009 orders**, **793 customers**, and an **Average Order Value of \$458.61**. The sales trend indicates overall growth over the four-year period, while the category and regional visuals highlight differences in sales and profitability across the business. These KPIs provide management with a concise view of overall business performance and areas requiring further analysis.

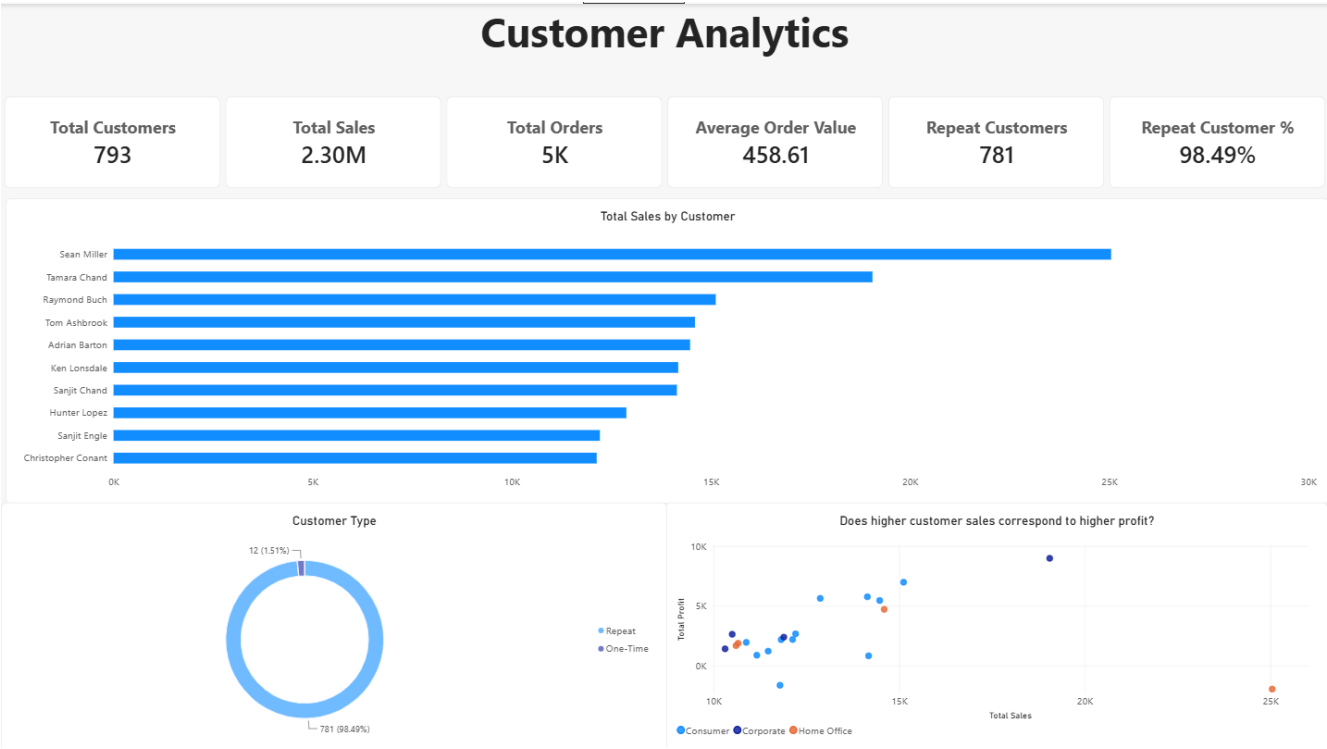
15.2 Product Performance Dashboard

This page focuses on top and bottom products, product profitability, and product-level performance.



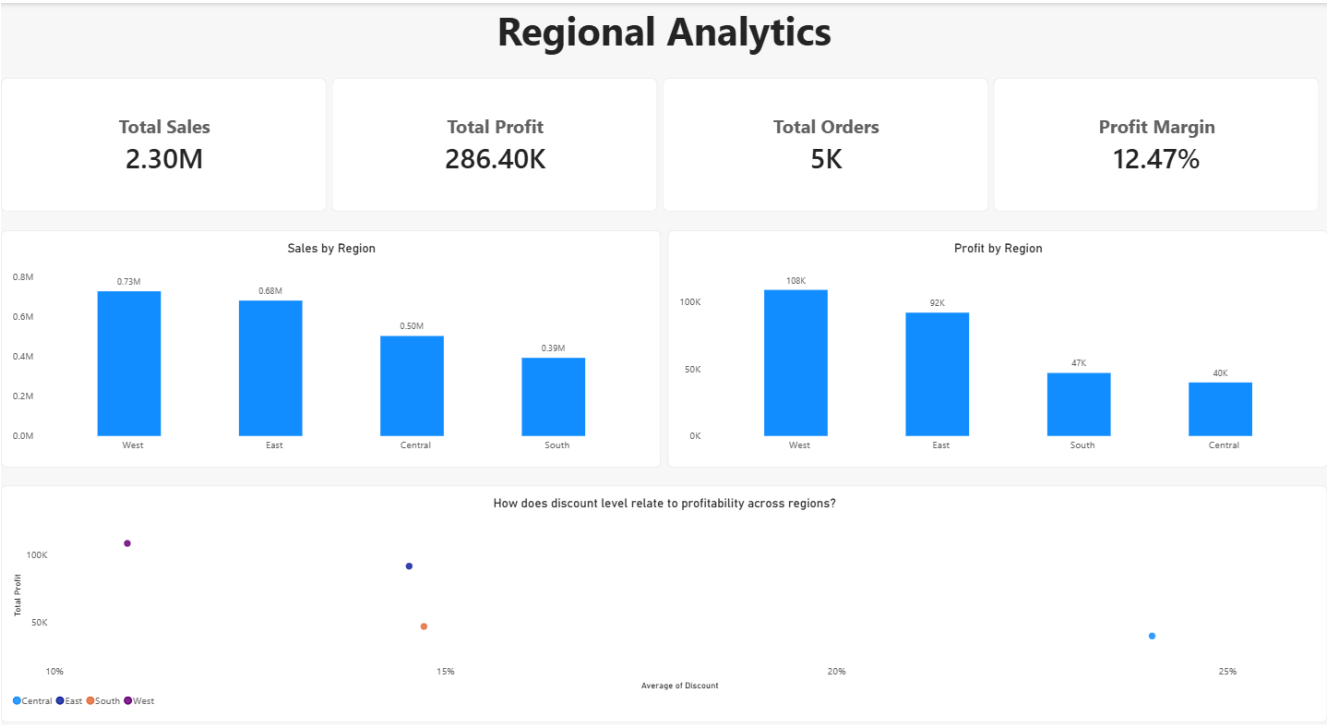
15.3 Customer Analytics Dashboard

This page presents customer value, top customers, repeat-customer information, and customer purchasing behavior.



15.4 Regional Analysis Dashboard

This page compares regional sales and profit and supports analysis of regional performance and discount-related profitability.



16. Key Business Findings

- Total sales were \$2.30 million with total profit of approximately \$286.40K and an overall profit margin of 12.47%.
- Technology generated the highest category sales and profit, while Furniture had a substantially lower profit margin of about 2.49%.
- The West region recorded the highest sales and profit, while Central had the lowest profit margin among the regions.
- The highest-selling product was also highly profitable, but several other high-sales products generated low or negative profit.
- High discount levels were associated with negative aggregate profitability in several discount groups.
- The customer base contained 781 repeat customers and 12 one-time customers based on the analysis.

17. Business Problems Identified

- Low profitability in Furniture despite meaningful sales volume.
- Several individual products generated negative profit.
- Some high-sales customers generated relatively low or negative profit.
- High discount levels were associated with negative aggregate profit.
- Regional profitability varied considerably, with Central showing a lower margin than the other regions.

18. Business Opportunities

- Improve product-level profitability monitoring rather than focusing only on sales.
- Review pricing and discount policies for high-discount products.
- Investigate the product mix and profitability drivers in Furniture.
- Use customer profitability alongside customer sales when evaluating customer value.
- Study regional differences to identify practices that could be applied across lower-margin regions.

19. Recommendations

- Review products with repeated negative profit and assess pricing, discounting, and product-level economics.
- Monitor discount levels and establish management review thresholds for high-discount transactions.
- Evaluate Furniture products and sub-categories with low margins to identify improvement opportunities.
- Use both sales and profit metrics when prioritizing customers and products.
- Use regional sales and profit dashboards regularly to monitor changes in regional performance.

20. Limitations

- The dataset covers 2014–2017, so the findings describe the historical dataset and may not represent current market conditions.
- The available fields do not provide detailed cost components, so exact drivers of product-level profit cannot be decomposed.
- Correlation analysis identifies relationships between variables but does not establish causal effects.
- Discount analysis is based on aggregated historical observations and may be affected by product mix, region, customer type, and other factors.

- Customer behavior analysis is limited to the variables available in the dataset.

21. Conclusion

The project transformed a raw e-commerce dataset into a structured analytical workflow covering data understanding, cleaning, Excel analysis, SQL analysis, Python/Pandas EDA, statistical analysis, and Power BI reporting. The results demonstrate that sales volume alone does not fully describe business performance: profitability varies by category, region, product, customer, and discount level.

The Power BI dashboard provides management with an interactive way to monitor the major KPIs and investigate the analytical areas identified in this report. The findings and recommendations provide a data-supported basis for reviewing product profitability, discount practices, customer value, and regional performance.

22. Project Deliverables

Folder	Deliverable
01_Project_Proposal	Business Requirement Document
02_Raw_Data	Raw e-commerce dataset
03_Cleaned_Data	Cleaned dataset and Cleaning Log
04_Excel_Analysis	Excel Analysis workbook
05_SQL	PostgreSQL business analysis SQL
06_Python	Data Exploration, Data Cleaning and EDA notebooks
07_Visualization	EDA and statistical charts
08_PowerBI	Power BI dashboard (.pbix)
09_Report	Final Business Analytics Report
10_Presentation	Final presentation

